

Venkat Sai Bhushan Mandava

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SUMMARY

Data Analyst with **2+ years of experience** designing and deploying robust **data pipelines** and **ETL workflows** on **cloud platforms**. Proficient in **SQL programming** and **data transformation**, with a track record of enhancing **data quality** for **analytical insights**. Demonstrates strong **cross-functional collaboration** to build **scalable solutions** that integrate **predictive analytics** and **operational intelligence**.

WORK EXPERIENCE

Samsung | Machine Learning Engineer | Intern

Jul 2021 - Mar 2022

- Developed and deployed **CNN-based deep learning models** using **PyTorch** and **TensorFlow** to enhance **semiconductor defect detection**, contributing to a **10% reduction in defects** and supporting **predictive analytics initiatives** while collaborating with **cross-functional teams**.
- Designed and implemented robust, automated **ETL pipelines** for **data cleaning** and **preprocessing**, leveraging **SQL** and **data transformation best practices** to ensure high **data quality** for **classification** and **anomaly detection models**.
- Applied **K-means clustering** to segment over **10,000 semiconductor defect data points**, collaborating with **engineering** and **product teams** to extract **actionable insights** that improved **production efficiency** and reduced **errors by 12%**.

KLUniversity | Peer Mentor | Part-Time

Jul 2020 - Jun 2021

- Guided a **student team** through a **machine learning project**, simplifying complex concepts such as **model training** and **feature engineering** while demonstrating practical **data analysis techniques** using **Python** and **Excel**, fostering **data-driven decision-making**.
- Revamped the **data mining syllabus** in collaboration with **faculty** to integrate **industry trends**, emphasizing advanced **data analysis methods** and **statistical programming**, and developed sample **reports** and **dashboards** to illustrate actionable **analytical insights**.

PROJECTS

Virtual Assistant for Drafting Emails | [GithubRepo](#) | [Blog](#)

- Engineered a **chaining-pattern virtual assistant** in **Python** using **asyncio**, **LLM-based multi-agent architecture**, and **Pydantic** for structured planning and execution of personalized emails via the **Gmail API**. Integrated **custom toolchains**, **OAuth2 authentication**, and real-time **streaming with Runner** to fully automate outreach from request.txt prompts

Job Search Agent | [GithubRepo](#) | [Blog](#)

- Built **Job Search AI** using **Python**, **Tavily API**, and **AI models (LLM)** to automate job searches, evaluate descriptions, and send **email alerts**. Leveraged **user preferences** and **environment variables** for secure and customizable automation. Showcased skills in **API integration**, **AI-driven insights**, and **automation** for optimized job-hunting.

German2English Seq2Seq | [GithubRepo](#) | [Blog](#) | [Reference](#)

- Built a **Seq2Seq** machine translation model using **PyTorch**, **LSTMs**, teacher forcing, and **BLEU** evaluation on the Multi30k dataset. Engineered data preprocessing, tokenization, vocabulary creation, and model training with **CrossEntropyLoss** and **Adam optimizer**.

Neurovision | [GithubRepo](#) | [Blog](#)

- Built a brain tumor detection system, experimenting with **VGG16** and **RCNN** before optimizing with **YOLOv8** for real-time, high-accuracy performance. Integrated **preprocessing**, **transfer learning**, **Grad-CAM** visualizations, and deployed a **Streamlit** app with confidence scores. Enabled model tracking and experiment logging via **Weights & Biases** for robust monitoring and reproducibility.

MLOps-Based Fraud Detection Deployment | [GithubRepo](#) | [Blog](#)

- Developed an end-to-end **MLOps** pipeline with **FastAPI** for real-time model serving, **Docker** for containerization, and **Weights & Biases** (W&B) for model versioning, experiment tracking, and performance monitoring, along with **MLflow** for tracking models.

Yelp-Camp | [GithubRepo](#) | [Blog](#)

- Developed a **CI/CD** pipeline for the Campground app using **Jenkins**, **Docker**, **SonarQube**, and **AWS EKS**, automating deployment and ensuring code quality. This project highlights my expertise in **DevOps**, cloud infrastructure, and software deployment.

EDUCATION

University of New Haven, New Haven, Connecticut

Aug 2023 - Present

Master's In Computer Science (GPA:3.4)

SKILLS

- **Machine Learning, Data Engineering & Analytics:** Machine Learning, Deep Learning, Neural Networks, NLP, GenerativeAI, Data Preprocessing, Feature Engineering, Model Fine-Tuning, SQL, MLOps, Predictive Models, Data Engineering, DBT Programming, ETL Jobs, Data Transformation Skills, Analytical Insights, Data Visualization Tools
- **Frameworks & Libraries:** TensorFlow, PyTorch, Keras, Django, Scikit-Learn, OpenCV
- **Languages:** C, C++, JavaScript, Java, Python
- **Databases:** MongoDB, SQL
- **Cloud Platforms:** Azure, AWS, GCP, AWS SageMaker
- **Containerization, Orchestration & System:** Docker, Kubernetes, Distributed System
- **MLOps Tools:** MLflow, Weights & Biases, Git, CI/CD Pipelines